

8(a)(i)	energy required to separate the nucleons (in the nucleus)	M1
	to infinity	A1
8(a)(ii)	curve starting close to the origin and forming a single peak	B1
	peak shown to left of centre, with steep line on LHS of peak and shallow line on RHS of peak	B1
8(b)(i)	fusion	B1
8(b)(ii)	both particles have low A values or both particles are at left-hand end of graph	B1
	He-3 has higher binding energy (per nucleon) than H-2	B1
8(c)	$\Delta m = [(2 \times 2.014102) - (3.016029 + 1.008665)] \text{ u}$ (= 0.00351 u)	C1
	$E = \Delta mc^2$	C1
	$= 0.00351 \times 1.66 \times 10^{-27} \times (3.00 \times 10^8)^2$ (= $5.24 \times 10^{-13} \text{ J}$)	C1
	1.00 mol of deuterium forms 0.500 mol of helium-3	C1
	total energy = $0.500 \times 6.02 \times 10^{23} \times 5.24 \times 10^{-13}$ = $1.58 \times 10^{11} \text{ J}$	A1